

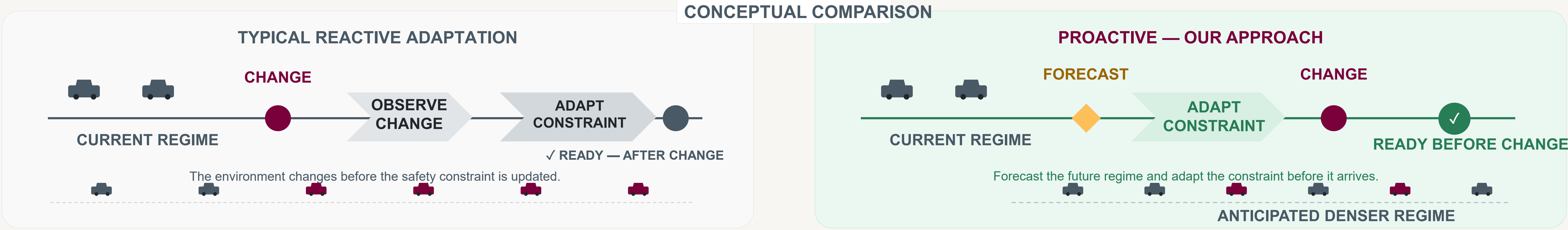
PROACTIVE CONTEXT-FORECASTED SAFETY CONSTRAINTS FOR NONSTATIONARY REINFORCEMENT LEARNING

Tim Tomashevskiy
Department of Computing and Software · McMaster University



WHY PROACTIVE SAFETY?

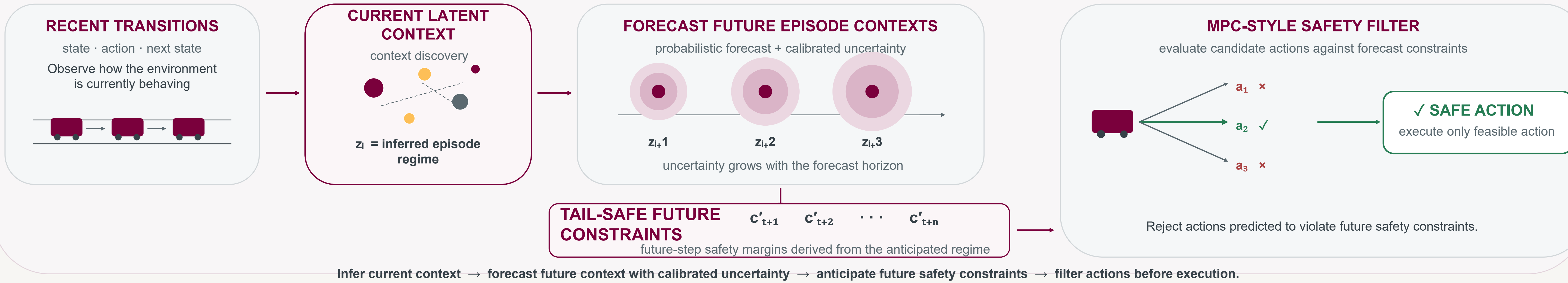
Safety constraints designed for one operating regime can become outdated or overly conservative as the environment changes.



SAFETY CONSTRAINTS SHOULD ANTICIPATE CHANGE — NOT MERELY REACT TO IT.

HOW DOES IT WORK?

OBSERVE → UNDERSTAND → ANTICIPATE → ACT
from experience infer the regime forecast risk filter before execution

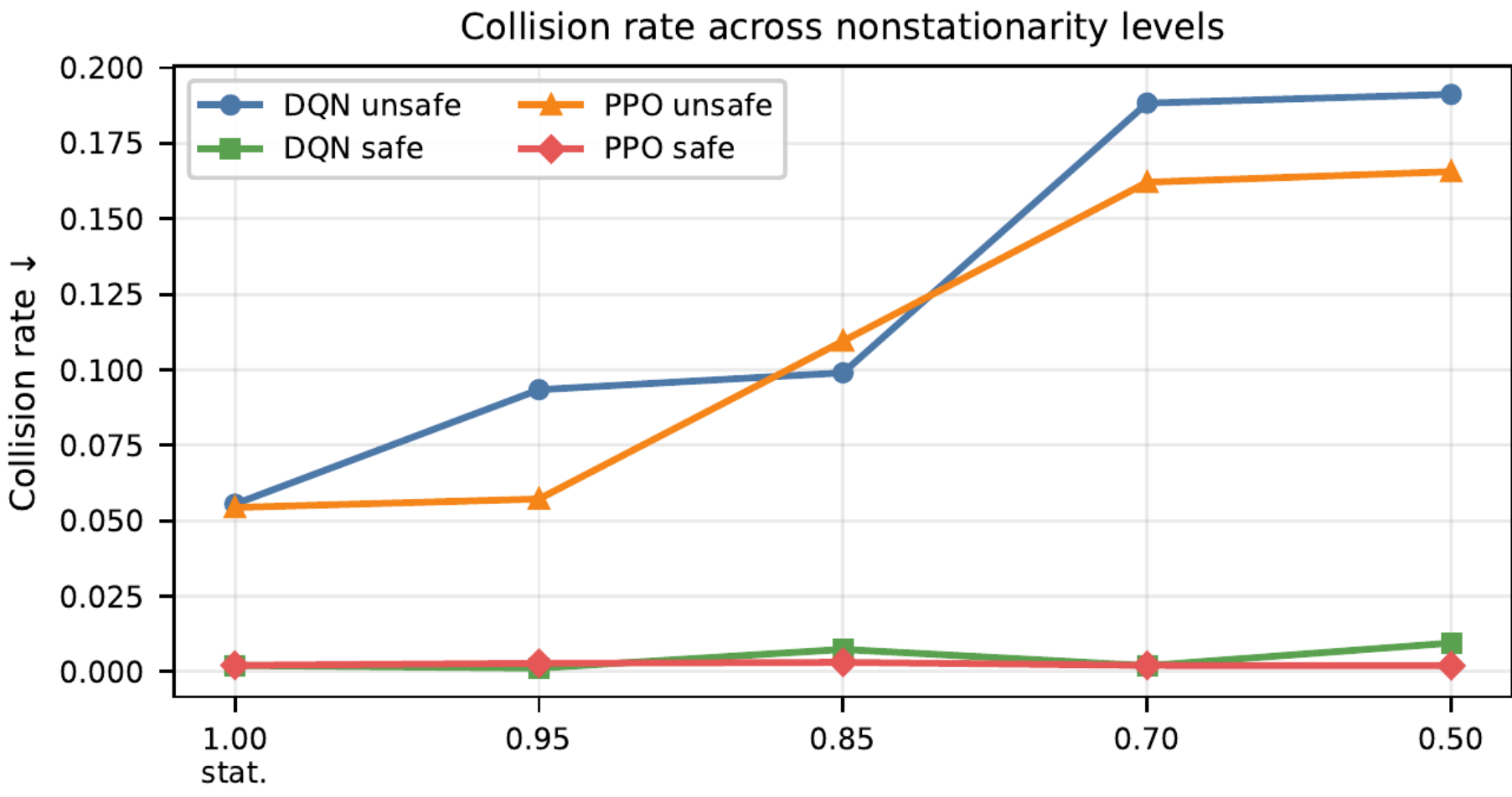


EXPERIMENTAL EVIDENCE

DOES IT WORK?

98.9% COLLISION REDUCTION
18.83% → 0.20%

DQN · strong nonstationarity · $p_{\text{stay}} = 0.70$

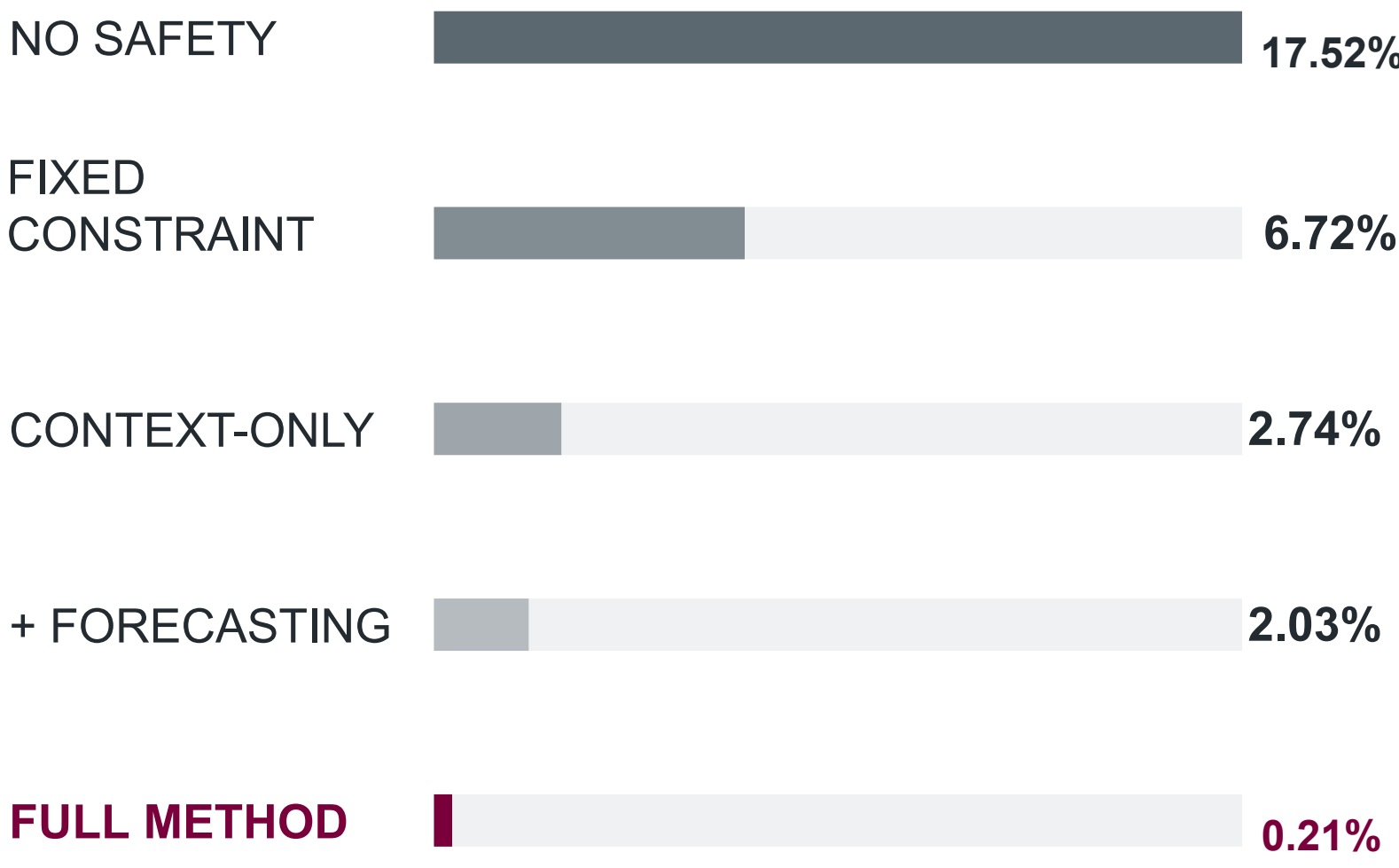


PPO: 16.21% → 0.21%
Same strong-nonstationarity setting

>90% reduction across every evaluated setting
usually >97%

Unsafe policies deteriorate as context switching increases;
the safety-enabled curves remain near zero.

WHICH COMPONENTS MATTER?



17.52 → 6.72 → 2.74 → 2.03 → 0.21%
No safety → Fixed → Context → Forecast → Full

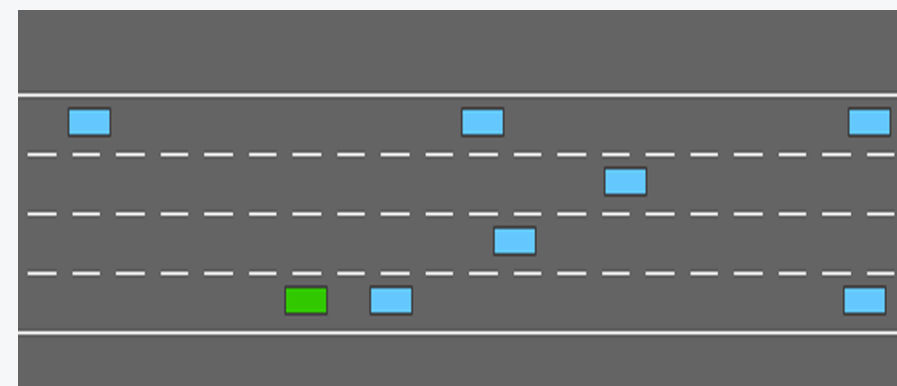
Static safety helps; context dependence and forecasting progressively reduce collisions further.

Component analysis at $p_{\text{stay}} = 0.70$;
aggregated across DQN and PPO.

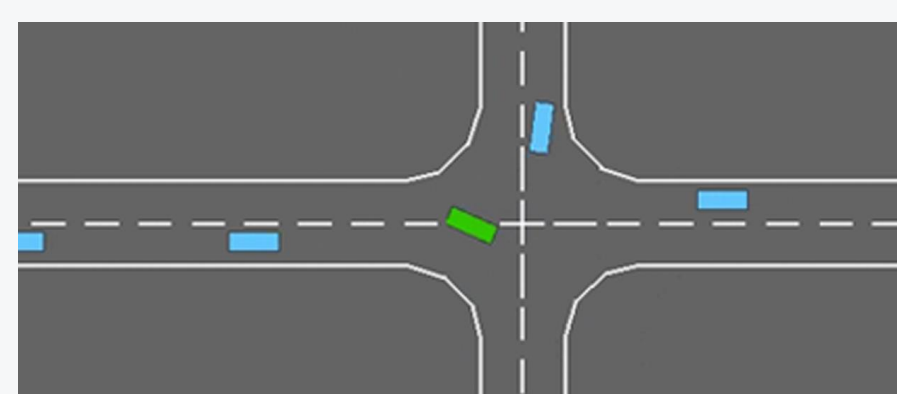
FULL METHOD · 0.21% COLLISIONS

HELD-OUT LAYOUT STRESS TEST

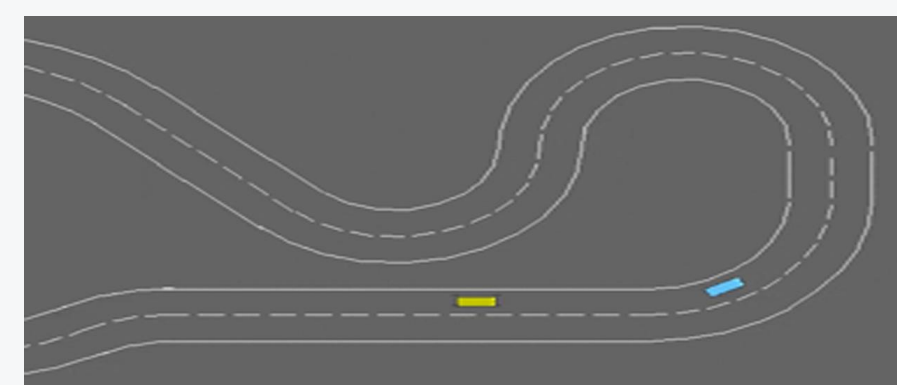
Representative HighwayEnv environment views



HIGHWAY-v0
97.1% ↓
collision reduction



INTERSECTION-v0
92.0% ↓
collision reduction



RACETRACK-v0
95.7% ↓
collision reduction

Collision reductions persist across all evaluated held-out layouts.

Percentages aggregate DQN/PPO across
evaluated nonstationarity levels.

Representative views: HighwayEnv / Farama Foundation

WHAT SAFETY CLAIM DO WE MAKE?

CONDITIONAL HORIZON-LEVEL SAFETY

$$\Pr(\exists k \leq n : d_{t+k} < \epsilon) \leq \alpha + nq$$

Probability of a clearance violation within the forecast horizon is bounded by $\alpha + nq$.

CALIBRATED CONTEXT COVERAGE
 $1 - \alpha$

TAIL-SAFE CLEARANCE PREDICTION
 q

CONTROL-AWARE MPC FILTERING

HORIZON-LEVEL SAFETY BOUND $\alpha + nq$

Conditional probabilistic guarantee over the forecast horizon — not unconditional hard safety.
Requires calibrated context coverage, tail-prediction coverage, and MPC feasibility.

TAKE-HOME

Proactive context forecasting enables safety constraints to adapt before anticipated environmental change.

FORECAST THE CONTEXT.
ADAPT THE CONSTRAINT.
ACT BEFORE THE RISK ARRIVES.



SCAN FOR PAPER

tomashet@mcmaster.ca

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